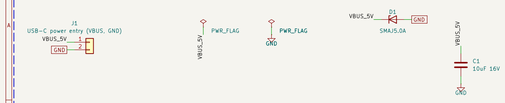
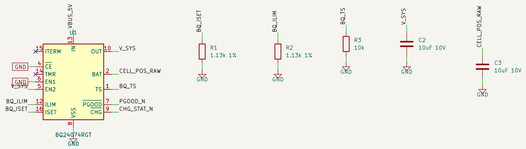
A — USB-C power input (5 V only)

VBUS and GND only; D+/D−/CC belong to the Servo Board. No PD, no 9 V.
SMAJ5.0A unidirectional TVS (drawn as Zener: cathode on VBUS). IN cap 1–10 uF per datasheet.



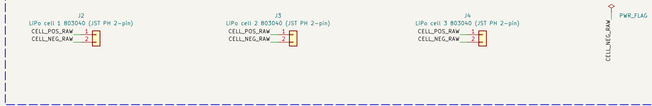
B — BQ24074 charger / power path

ICHG = 890 AOhm / 1.13k = 0.79 A, IIN(max) = 1610 AOhm / 1.13k = 1.42 A (RILIM >= 1.1k).
Requires a USB-C source rated >= 1.5 A. ENZ=OUT, EN1=GND: ILIM-programmed. CE=GND.
TS: 10k fixed (cells have no NTC). TMR, ITERM open: default timers, 10 % termination.



C — 1S3P pack (3 x 803040 in parallel)

Voltage-match cells within 30–50 mV before first parallel connection.



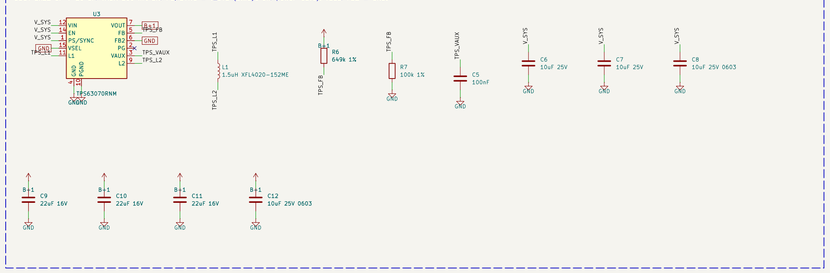
D — Pack protection (DW01A + FS8205A)

Datasheet reference: R 470 Ohm + 100 nF on VCC, 2k on CS. OD -> G1 (discharge FET, S1 at cell negative); OC -> G2 (charge FET, S2 at GND). Raw cell negative reaches GND only through Q1.



E — B+1 regulator (TPS63070, 6.0 V)

VOUT = 0.8 V x (1 + 649k/100k) = 5.99 V. Datasheet Fig. 10: 1.5 uH, CIN 2x10 uF + 10 uF, COUT 3x22 uF + 10 uF, VAUX 100 nF. EN, PS/SYNC = V_SYS (always on, power-save). VSEL, FB2 = GND.



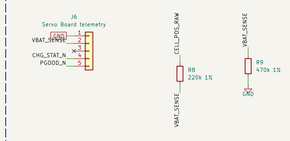
F — Machine power output

B+1 and GND to the former battery-terminal pads; S901 remains the master switch.



G — Telemetry to Servo Board

VBAT_SENSE = 4.2 V x 470k/690k = 2.86 V max; always-on divider (6 uA).
VBAT_SENSE_EN (pin 3) not used on this board. CHG/PGOOD are pulled up on the Servo Board.



Test points (guide 12)



Connector families/pitch not yet chosen: generic connectors, empty footprints
USB-C 5 V only (no PD). 1S3P 803040 LiPo. B+1 = 6.0 V into the former battery pads.
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wm06c-servo (CERN-OHL-P v2)	
Sheet: /	
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Size: D	Date: 2026-11-09 — **Rev: A-draft**
KiCad E.D.A. 10.0.8	Id: 1/1